

Assignment 4 - Answer Sheet

Sheet: All Errors

File	Line	Error Type	Description	Category
1.c	11	Syntax	Missing semicolon after 'int department = 1'	Syntax
1.c	12	Semantic/Type	int marks = "90" — assigning string literal to int	Semantic
1.c	26	Logical	if(x = 10) — assignment instead of comparison (==)	Logical
1.c	31	Scope	'score' is undeclared (no such variable exists)	Scope
1.c	33	Scope/Shadowing	int totalStudents = 100 shadows the global variable	Scope
1.c	50	Linker	add(a,b) — implicit declaration, no prototype before call	Linker
1.c	55,58	Scope	'temp' declared in inner block, used outside it	Scope
1.c	61	Runtime/Array	arr[10] = 100 — array index out of bounds (arr[5])	Runtime
1.c	66	Semantic/Format	printf("%d", salary) — float passed with %d format	Semantic
1.c	84,87	Scope	'result' declared in for-loop block, used outside	Scope
1.c	90-93	Logical	if and else both print 'Eligible' — same output	Logical
1.c	98	Semantic/Function	displayData() called with 0 args, but needs 1 (int)	Semantic
1.c	129	Runtime/Logic	while(z > 0); — semicolon makes infinite empty loop	Runtime
1.c	134	Semantic/Type	char grade = "A" — string literal to char	Semantic
1.c	138	Semantic/Format	printf("%s", number) — int passed with %s format	Semantic
1.c	141-142	Runtime/Pointer	*ptr = 100 where ptr is NULL — null pointer dereference	Runtime
1.c	146	Linker	calculateAverage(10,20) — called but never defined	Linker
2.c	10	Semantic/Type	int marks = "90" — string literal assigned to int	Semantic
2.c	16	Syntax	printf(...) missing semicolon at end of statement	Syntax
2.c	22	Linker	extern int total — declared but never defined in any file	Linker
2.c	36	Linker	add(10,20) — implicit declaration, defined later without prototype	Linker
2.c	42	Semantic/Register	&number — cannot take address of register variable	Semantic
2.c	44	Semantic/Type	static int salary = "50000" — string literal to int, not compile-time constant	Semantic
2.c	49,52	Scope	'department' declared in inner block, used outside it	Scope
2.c	58,61	Scope	'temp' declared in for-loop, used outside it	Scope
2.c	63	Runtime/Array	arr[10] = 500 — out of bounds (arr[5])	Runtime
2.c	67	Runtime/Arithmetic	a/b where b=0 — division by zero	Runtime
2.c	74	Logical	if(value = 100) — assignment instead of comparison	Logical
2.c	80-83	Logical	if and else both print 'Pass' — same output for both branches	Logical
2.c	96	Semantic/Type	char grade = "A" — string literal to char	Semantic
2.c	106	Logical	printf("%d", data) — data is uninitialized (garbage value)	Logical
2.c	122-123	Runtime/Pointer	*ptr = 100 where ptr is NULL — null pointer dereference	Runtime
2.c	130	Linker	calculateAverage(10,20) — called but never defined	Linker

File	Line	Error Type	Description	Category
2.c	134	Semantic/Type	int employeeId = "EMP101" — string literal to int	Semantic
2.c	138	Runtime/Logic	while(counter > 0); — semicolon makes infinite empty loop	Runtime
2.c	148	Linker	extern int companyId — declared but never defined	Linker
3.c	25	Syntax	int choice — missing semicolon	Syntax
3.c	59	Syntax	default — missing colon (:) in switch statement	Syntax
3.c	91	Logical	if(index > 5) — check should be >=5 and before increment	Logical
3.c	103	Logical/Array	displayStudents() loop: i<=totalBooks — off-by-one error	Logical
3.c	119	Runtime/Arithmetic	return total / 0 — division by zero	Runtime
3.c	124,125	Semantic/Type	int highest = students[0].marks — float assigned to int	Semantic
3.c	148	Scope	totalStudent — undeclared (should be totalStudents)	Scope
3.c	150	Linker	printGrade() — called but never defined	Linker
3.c	152	Linker	calculateRank() — called but never defined	Linker
3.c	159	Semantic/Type	s.rollNo = "101" — string literal to int	Semantic
3.c	163	Semantic/Type	s.marks = "90" — string literal to float	Semantic
3.c	175	Logical	students[roll].marks = 100 — uses rollNo as array index directly	Logical
3.c	185	Semantic/Type	students[roll] = NULL — cannot assign NULL to struct	Semantic
3.c	193	Logical	if(name == students[0].name) — string compare with == instead of strcmp()	Logical
3.c	201	Semantic/Type	avg = calculateAverage() — float returned, assigned to int	Semantic
3.c	212	Logical/File	fopen('student.txt','r') then fprintf() — read mode but writing	Logical
3.c	220	Logical/File	fopen('missing.txt','r') — no error checking for missing file	Logical
4.c	68	Logical	if(totalAccounts > 10) — check should be >=10 and before increment	Logical
4.c	79	Logical/Array	displayAccounts() loop: i<=totalAccounts — off-by-one error	Logical
4.c	95,110	Logical	accounts[acc] used directly as index — should search by accNo	Logical
4.c	115	Logical	if(balance > amount) — wrong condition; withdraws when balance is LESS	Logical
4.c	192	Scope	totalAccount — undeclared (should be totalAccounts)	Scope
4.c	194	Linker	calculateInterest() — called but never defined	Linker
4.c	196,252	Linker/Type	saveAccounts() — called before defined, implicit declaration conflict	Linker
4.c	203	Semantic/Type	a.accNo = "1001" — string literal to int	Semantic
4.c	207	Semantic/Type	a.balance = "50000" — string literal to float	Semantic
4.c	228	Semantic/Type	accounts[accNo] = NULL — cannot assign NULL to struct	Semantic
4.c	236	Logical	if(name == accounts[0].name) — string compare with == instead of strcmp()	Logical
4.c	247	Linker	generateAverage() — called but never defined	Linker
4.c	256	Logical/File	fopen('accounts.txt','r') then fprintf() — read mode but writing	Logical
4.c	264	Logical/File	fopen('missing.txt','r') — no error checking	Logical
4.c	309	Runtime/Array	arr[10] = 500 — out of bounds (arr[5])	Runtime
4.c	311-312	Runtime/Pointer	*ptr = 100 where ptr is NULL — null pointer dereference	Runtime

File	Line	Error Type	Description	Category
4.c	314	Runtime/Arithmetic	a/b where b=0 — division by zero	Runtime
5.c	27	Syntax	int choice — missing semicolon	Syntax
5.c	65	Syntax	default — missing colon (:) in switch statement	Syntax
5.c	83	Logical	if(totalBooks > 20) — check should be >=20 and before increment	Logical
5.c	94	Logical/Array	displayBooks() loop: i<=totalBooks — off-by-one error	Logical
5.c	112,122	Logical	books[id] used directly as index — should search by bookId	Logical
5.c	173	Scope	totalBook — undeclared (should be totalBooks)	Scope
5.c	175,235	Linker/Type	saveLibrary() — called before defined, implicit declaration conflict	Linker
5.c	177	Linker	calculateFine() — called but never defined	Linker
5.c	184	Semantic/Type	b.bookId = "101" — string literal to int	Semantic
5.c	190	Semantic/Type	b.price = "500" — string literal to float	Semantic
5.c	198	Logical	if(title == books[0].title) — string compare with == instead of strcmp()	Logical
5.c	213	Semantic/Type	books[id] = NULL — cannot assign NULL to struct	Semantic
5.c	222	Logical	updateBook() uses books[id] directly as index — should search by bookId	Logical
5.c	230	Linker	calculateAveragePrice() — called but never defined	Linker
5.c	240	Logical/File	fopen('library.txt','r') then fprintf() — read mode but writing	Logical
5.c	248	Logical/File	fopen('missing.txt','r') — no error checking	Logical
5.c	275-276	Runtime/Pointer	*ptr = 100 where ptr is NULL — null pointer dereference	Runtime
5.c	278	Runtime/Array	arr[10] = 200 — out of bounds (arr[5])	Runtime
5.c	280-281	Runtime/Arithmetic	a/b where b=0 — division by zero	Runtime
5.c	288	Runtime/Buffer	strcpy(name, 'Programming') into char[5] — buffer overflow	Runtime
5.c	296	Runtime/Memory	malloc(5) bytes used as array of 20 ints — massive overflow	Runtime
5.c	300	Runtime/Memory	ptr[0] after free(ptr) — use-after-free	Runtime
5.c	302	Runtime/Memory	free(ptr) called twice — double free	Runtime

Sheet: Summary by File

File	Linker	Logical	Runtime	Scope	Semantic	Syntax
1.c	2	2	3	4	5	1
2.c	4	3	4	2	5	1
3.c	2	6	1	1	5	2
4.c	3	7	3	1	3	0
5.c	3	7	7	1	3	2

Sheet: Summary by Category

Category	Count
Linker	14

Category	Count
Logical	25
Runtime	18
Scope	9
Semantic	21
Syntax	6